

SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. - Information Science and Engineering	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: IS43	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

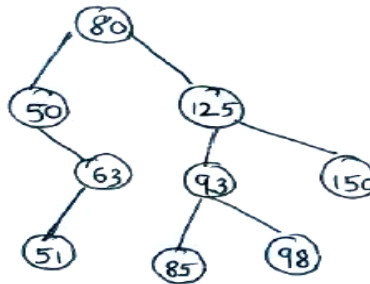
UNIT - I

- Develop an algorithm to determine the minimum and maximum values in an array of integers. (Here $n \geq 1$ and the entries in the array need not be distinct and determine its worst case time complexity). CO1 (07)
 - Obtain an algorithm for Tower of Hanoi puzzle and obtain the recurrence relation to find the total number of disk movements and solve the corresponding recurrence relation. CO1 (08)
 - Compare the orders of growth of $\log_2 n$ and $\sqrt{n}$. CO1 (05)
- Consider the following algorithm : CO1 (08)
 Algorithm Mystry(A[],n)
 for i=0 to n-2 do
 for j=i+1 to n-1 do
 if(a[i]=a[j])
 return 0;
 end if
 end for
 end for
 return 1;
 i)What does this algorithm compute?
 ii)What is its basic operation?
 iii)How many times is the basic operation executed?
 iv)What is the efficiency class of this algorithm?
 - Write the general plan for analyzing non recursive algorithm and list three examples for non-recursive algorithm. CO1 (06)
 - Express f(n) using Big-theta, Big Omega and Big-Oh for the following functions: CO1 (06)
 i). $f(n) = 6 * 2^n + n^2$ ii). $f(n) = 10n^3 + 5$.

UNIT - II

- Sort the list E, X, A, M, P, L, E in alphabetical order using Bubble sort and analyse the efficiency of Bubble sort algorithm. CO2 (06)
 - Write pseudo code for a divide-and-conquer algorithm for finding the position of the largest element in an array of n numbers. Set up and solve a recurrence relation for the number of key comparisons made by your algorithm. CO2 (06)
 - Write the algorithm to sort the give list of elements using Merge sort and derive its worst case efficiency. CO2 (08)
- Apply Quick sort to arrange the list 6,10,13,5,8,2,11,4,32,1 in ascending order. Derive it best case efficiency. CO2 (08)
 - Apply the Horspool algorithm to search the pattern ZIGZAG in the given text A_ZIG,A_ZAG_AGAIN_A_ZIGZAG. CO2 (06)

- c) Traverse the following Binary Search Tree in preorder, inorder and postorder. Analyze the efficiency of preorder traversal algorithm. CO2 (06)



UNIT - III

5. a) Apply the dynamic programming algorithm to the following instance of the knapsack problem. Consider capacity $w=5$. CO3 (10)

Item	Weight	Value
1	2	\$10
2	1	\$12
3	3	\$22
4	2	\$15

Write the memory function knapsack algorithm.

- b) Implement Kruskal's algorithm for the graph shown Fig. 5(b), and find the minimum spanning tree. CO3 (10)

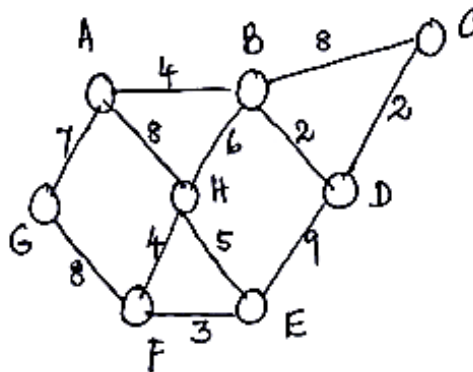


Fig. 5(b)

6. a) Define transitive closure of a graph. Apply Floyd's algorithm for the graph shown in Fig. 6(a). CO3 (10)

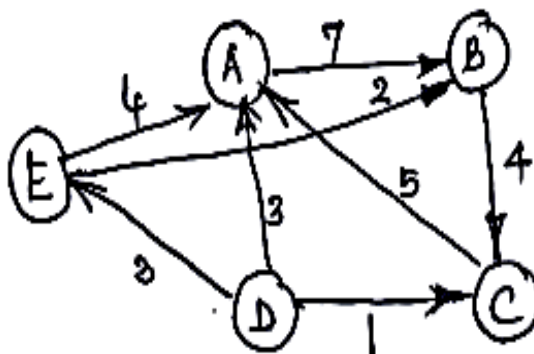


Fig. 6(a)

- b) Apply Dijkstra's algorithm, by assuming B as the source for the graph shown in Fig. 6(b). What is the real world application of this algorithm? CO3 (10)

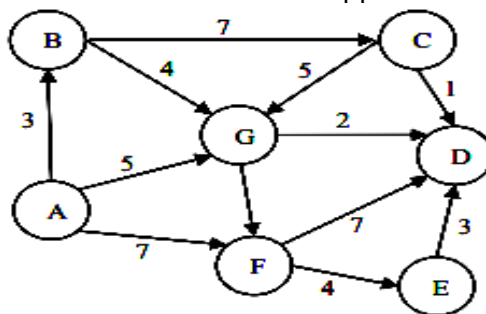
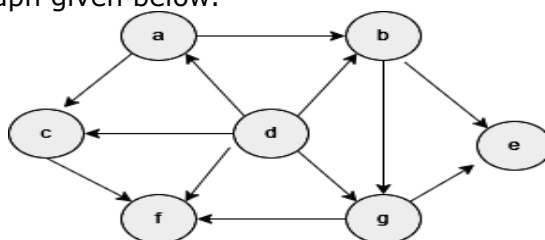


Fig. 6(b)

UNIT- IV

7. a) Show the working of insertion sort algorithm for the given input. CO4 (08)
87, 48, 63, 95, 28, 32, 16.
- b) What is the notion of a heap? What are the conditions that needs to be satisfied. List the properties of heap. CO4 (06)
- c) Construct a 2-3 tree for the list 9,5,8,3,2,4,7. CO4 (06)
8. a) What is the advantage of an AVL tree over a binary search tree? What is the balance requirement of AVL trees? CO4 (05)
- b) Write an algorithm that implements a depth-first search traversal of a given graph. CO4 (06)
- c) Apply the source elimination graph to represent vertices in topological order for the graph given below. CO4 (09)

**UNIT - V**

9. a) Explain how backtracking design technique solves difficult combinatorial problems. Draw the state-space tree for solving 4-queens problem using backtracking. CO5 (10)
 - b) Solve the following Traveling sales person problem to find the shortest tour for the given cost adjacency matrix using branch and bound technique: CO5 (10)
- | | | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| 1 | 0 | 1 | 2 | 5 |
| 2 | 6 | 0 | 4 | 3 |
| 3 | 5 | 3 | 0 | 7 |
| 4 | 1 | 4 | 3 | 0 |
10. a) Write a brief note on P, NP, and NP complete problems with one example for each. CO5 (10)
 - b) Differentiate between back tracking and branch-and-bound design techniques. CO5 (05)
 - c) Explain with examples, the need of approximation problems. CO5 (05)
